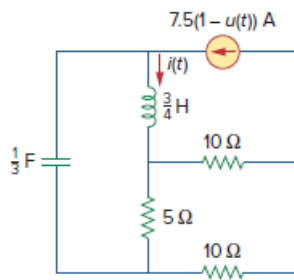


Problem

Refer to the circuit in Fig. 16.59. Calculate $i(t)$ for $t > 0$.

Figure 16.59:



Step-by-step solution

Step 1 of 9

Refer to circuit diagram in Figure 16.59 in the text book.

Represent the components of Figure 16.59 in s-domain.

$$\frac{1}{3} \text{ F} \Rightarrow \frac{1}{s \left(\frac{1}{3} \right)}$$

$$= \frac{3}{s}$$

$$\frac{3}{4} \text{ H} \Rightarrow sL$$

$$= \frac{3}{4}s$$

The unit step function is defined as follows.

$$u(t) = \begin{cases} 1 & \text{for } t \geq 0 \\ 0 & \text{for otherwise} \end{cases}$$

Hence,

$$7.5[1 - u(t)] = \begin{cases} 7.5(1 - 0) = 7.5 & \text{for } t < 0 \\ 7.5(1 - 1) = 0 & \text{for } t \geq 0 \end{cases}$$

Step 2 of 9

For $t < 0$, inductor is short circuited and capacitor is open circuited. The equivalent circuit is shown in Figure 1.

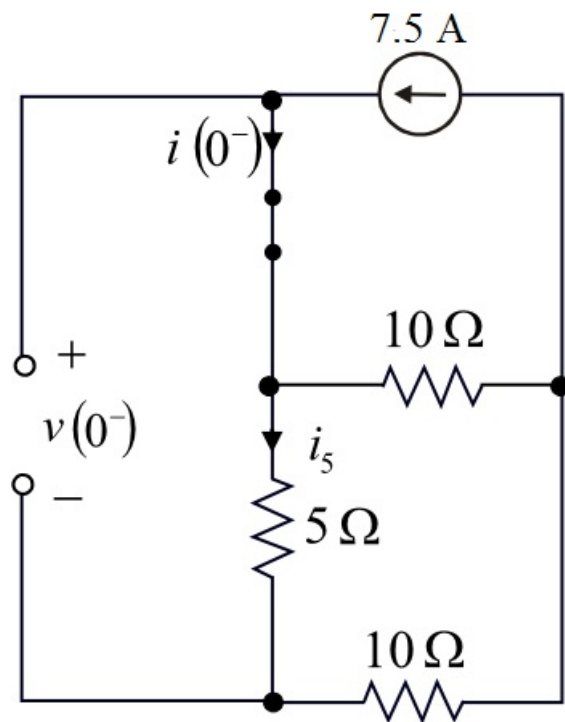


Figure 1

Step 3 of 9

The value of current $i(0^-)$ is,

$$i(0^-) = 7.5 \text{ A}$$

Inductor doesn't allow sudden changes in the current flowing through it.

$$\begin{aligned} i(0^-) &= i(0) \\ &= i(0^+) \\ &= 7.5 \text{ A} \end{aligned}$$

The equivalent voltage source for the initial value of the inductor is,

$$\begin{aligned} V_0(s) &= sL \times \frac{i_L(0)}{s} \\ &= \frac{3}{4} s \times \frac{7.5}{s} \\ &= \frac{22.5}{4} \end{aligned}$$

Step 4 of 9

The value of voltage $v(0^-)$ is,

$$\begin{aligned} v(0^-) &= 5(i_s) \\ &= 5 \left[7.5 \left(\frac{10}{10+5+10} \right) \right] \\ &= (37.5) \left(\frac{10}{25} \right) \\ &= 15 \text{ V} \end{aligned}$$

Capacitor doesn't allow sudden changes in the voltage across it.

$$\begin{aligned} v(0^-) &= v(0) \\ &= v(0^+) \\ &= 15 \text{ V} \end{aligned}$$

Step 5 of 9

The s-domain circuit for $t > 0$ is shown in Figure 2.

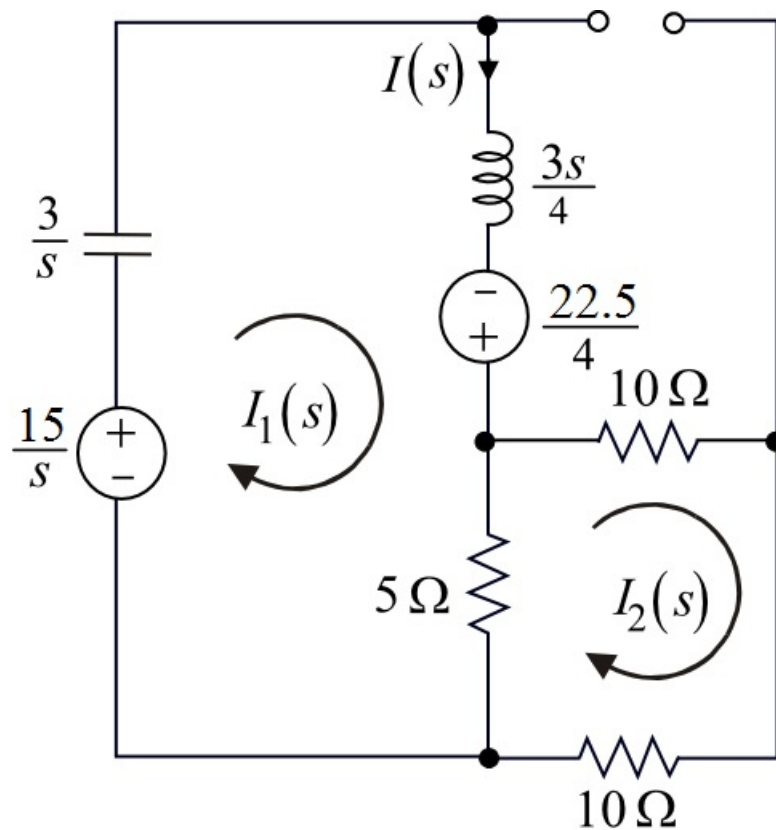


Figure 2

Step 6 of 9

Apply Kirchhoff's voltage law to Loop 1.

$$-\frac{15}{s} + \frac{3}{s}I_1(s) + \frac{3}{4}sI_1(s) - \frac{22.5}{4} + 5[I_1(s) - I_2(s)] = 0$$

$$\left(\frac{3}{s} + \frac{3}{4}s + 5\right)I_1(s) - 5I_2(s) = \frac{15}{s} + \frac{22.5}{4} \dots\dots (1)$$

Apply Kirchhoff's voltage law to Loop 2

$$10I_2(s) + 10I_2(s) + 5[I_2(s) - I_1(s)] = 0$$

$$25I_2(s) - 5I_1(s) = 0$$

$$I_2(s) = \frac{1}{5}I_1(s)$$

Step 7 of 9

Recall equation (1).

$$\left(\frac{3}{s} + \frac{3}{4}s + 5\right)I_1(s) - 5I_2(s) = \frac{15}{s} + \frac{22.5}{4}$$

Substitute $\frac{1}{5}I_1(s)$ for $I_2(s)$.

$$\left(\frac{3}{s} + \frac{3}{4}s + 5\right)I_1(s) - 5\left[\frac{1}{5}I_1(s)\right] = \frac{15}{s} + \frac{22.5}{4}$$

$$\left(\frac{12 + 3s^2 + 20s}{4s}\right)I_1(s) - I_1(s) = \frac{60 + 22.5s}{4s}$$

$$\left(\frac{12 + 3s^2 + 20s - 4s}{4s}\right)I_1(s) = \frac{60 + 22.5s}{4s}$$

$$I_1(s) = \frac{22.5s + 60}{3s^2 + 16s + 12}$$

$$I_1(s) = \frac{22.5s + 60}{(s + 0.9028)(s + 4.43)} \dots\dots (2)$$

Step 8 of 9

Apply partial fractions to simplify the current expression.

$$\frac{22.5s + 60}{(s + 0.9028)(s + 4.43)} = \frac{A}{s + 0.9028} + \frac{B}{s + 4.43} \dots\dots (3)$$

$$22.5s + 60 = A(s + 4.43) + B(s + 0.9028) \dots\dots (4)$$

Equate s coefficients from equation (4).

$$22.5 = A + B$$

$$A = 22.5 - B \dots\dots (5)$$

Equate constant terms from equation (4).

$$60 = 4.43A + 0.9028B$$

Substitute $22.5 - B$ for A .

$$60 = (4.43)(22.5 - B) + 0.9028B$$

$$60 = 99.675 - 4.43B + 0.9028B$$

$$3.5272B = 39.675$$

$$B = 11.25$$

Substitute 11.25 for B in equation (5).

$$A = 22.5 - B$$

$$= 22.5 - 11.25$$

$$= 11.25$$

Recall equation (3).

$$I_1(s) = \frac{A}{s + 0.9028} + \frac{B}{s + 4.43}$$

Substitute 11.25 for A and 11.25 for B .

$$I_1(s) = \frac{11.25}{s + 0.9028} + \frac{11.25}{s + 4.43}$$

Step 9 of 9

Determine the expression for current through the inductor $i(t)$ for $t > 0$.

Clearly from Figure 2.

$$I(s) = I_1(s)$$

$$= \frac{11.25}{s + 0.9028} + \frac{11.25}{s + 4.43}$$

Apply inverse Laplace transform to the equation.

$$i(t) = 11.25e^{-0.9028t}u(t) + 11.25e^{-4.43t}u(t)$$

$$= 11.25[e^{-0.9028t} + e^{-4.43t}]u(t) \text{ A}$$

Therefore, the expression for current through the inductor $i(t)$ for $t > 0$ is

$$\boxed{11.25[e^{-0.9028t} + e^{-4.43t}]u(t) \text{ A}}.$$